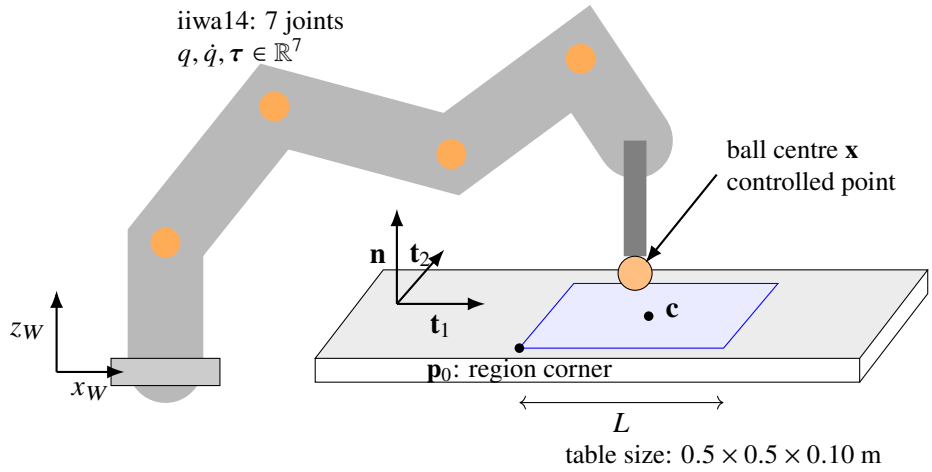
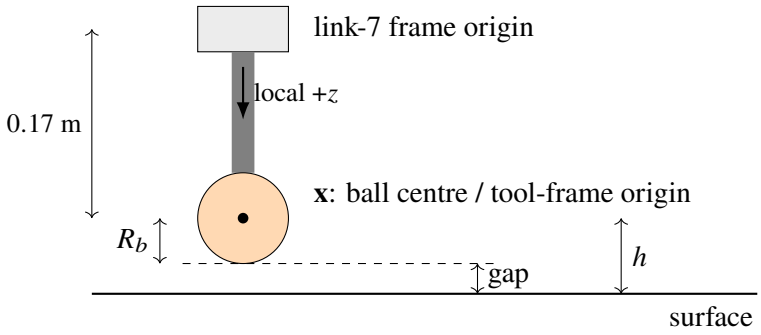
iiwa14 + Ball Tool + Table: Setup & Notation

Current master `f123d20`| Schematic, not to scale; flat table illustrated

A. Overall setup: robot, tool and wiping region



B. Tool detail: position and gap



Hover: gap > 0; aligned contact: gap near zero.

Parameters: ball radius 2 cm; tool mass 0.3 kg. The tool is rigidly welded to link 7, with no gripper. The connector is visual geometry only; the ball and table have collision geometry. The drawing illustrates connectivity, not an exact joint configuration.

Code:`wipe/sim.py:32 add_robot_and_tool`; `:56 table_pose`; `:62 add_table`; `wipe/config.py:38`.

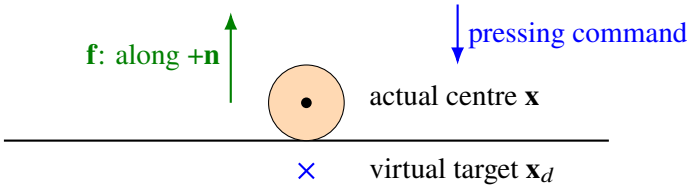
Notation: meaning, units and code

Notation	Meaning / units	Code reference
$\mathbf{x}$	Actual ball-centre position in world coordinates (m)	`sim.py:164 tool_state`
$\mathbf{x}_d$	Commanded position target (m); virtual point, not a physical object	`controllers.py:83 _calc`
R_b	Collision sphere radius, 0.02 m	`config.py:38 BALL_RADIUS`
$\mathbf{c}$	Region centre on the surface, not the box centre	`scene.py:72 make_scene`
$\mathbf{p}_0, L$	Region corner and side length, $L = 0.13$–0.17 m	Same as above
$\mathbf{n}$	Outward table normal, unit vector	`config.py:91 normal`
$\mathbf{t}_1, \mathbf{t}_2$	table-plane unit vectors	`scene.py:72 make_scene`
u_1, u_2, h	Ball-centre coordinates in the region frame (m)	`scene.py:58 to_plane`
θ	Table tilt about world x	`config.py:84 TableSpec`
$\mathbf{f}, f_n$	Table-on-tool force and normal component (N)	`controllers.py:90 ToolForceSensor`
K_n, k_e	Controller normal stiffness and effective contact stiffness (N/m)	`config.py:47,84`
K, D	Cartesian stiffness (N/m) and damping (N s/m) matrices	`config.py:59,63`
r_{wipe}	Virtual coverage radius, 0.02 m; distinct from physical radius R_b	`config.py:20 TaskParams`
u	Uncovered fraction; distinct from coordinates u_1, u_2	`coverage.py:44 uncovered`

Geometry: implemented in `scene.py:58--90`

$$\mathbf{n} = (0, -\sin\theta, \cos\theta)^T, \quad \mathbf{t}_1 = (1,0,0)^T, \quad \mathbf{t}_2 = \mathbf{n} \times \mathbf{t}_1,$$
$$\mathbf{p}_0 = \mathbf{c} - \tfrac{L}{2}\mathbf{t}_1 - \tfrac{L}{2}\mathbf{t}_2,$$
$$\mathbf{x} = \mathbf{p}_0 + u_1\mathbf{t}_1 + u_2\mathbf{t}_2 + h\mathbf{n}, \qquad \text{gap} = h - R_b.$$

Force and target: distinguish the directions



$$\mathbf{F}_{\text{cmd}} = K(\mathbf{x}_d - \mathbf{x}) - D\dot{\mathbf{x}}, \qquad f_n = \mathbf{n}^T\mathbf{f}.$$

$\mathbf{F}_{\text{cmd}}$ is the controller command; $\mathbf{f}$ is the actual contact force applied by the table to the tool. A target deeper than the actual centre produces a downward virtual-spring command. Default $K_n = 500$ N/m; k_e is selected by the table configuration. The sketch does not specify penetration depth.

Code: `controllers.py:83` computes commanded force; `:90` reads contact force. Height h and gap use the undeformed surface as reference; Drake computes penetration from collision geometry.